

EXPERIMENT-15

Q1. Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Algorithm

Main Algorithm

1. Start.

2. Declare integer variables num, temp, digit, choice and flag.

3. Input a positive integer num.

4. Display the menu:

1 – Check whether all digits are even.

2 – Check whether all digits are prime.

5. Input the user's choice.

6. Store num in temp.

7. Use switch(choice) to perform the selected operation.

8. Stop.

Case 1: Check Whether All Digits Are Even

1. Set flag = 1.
2. Repeat while temp > 0:
Find the last digit using digit = temp % 10.
Check whether digit % 2 != 0.
If true, set flag = 0 and break the loop.
Otherwise, remove the last digit using temp = temp / 10.
3. If flag == 1, display "All digits are even".
4. Otherwise, display "All digits are not even".

Case 2: Check Whether All Digits Are Prime

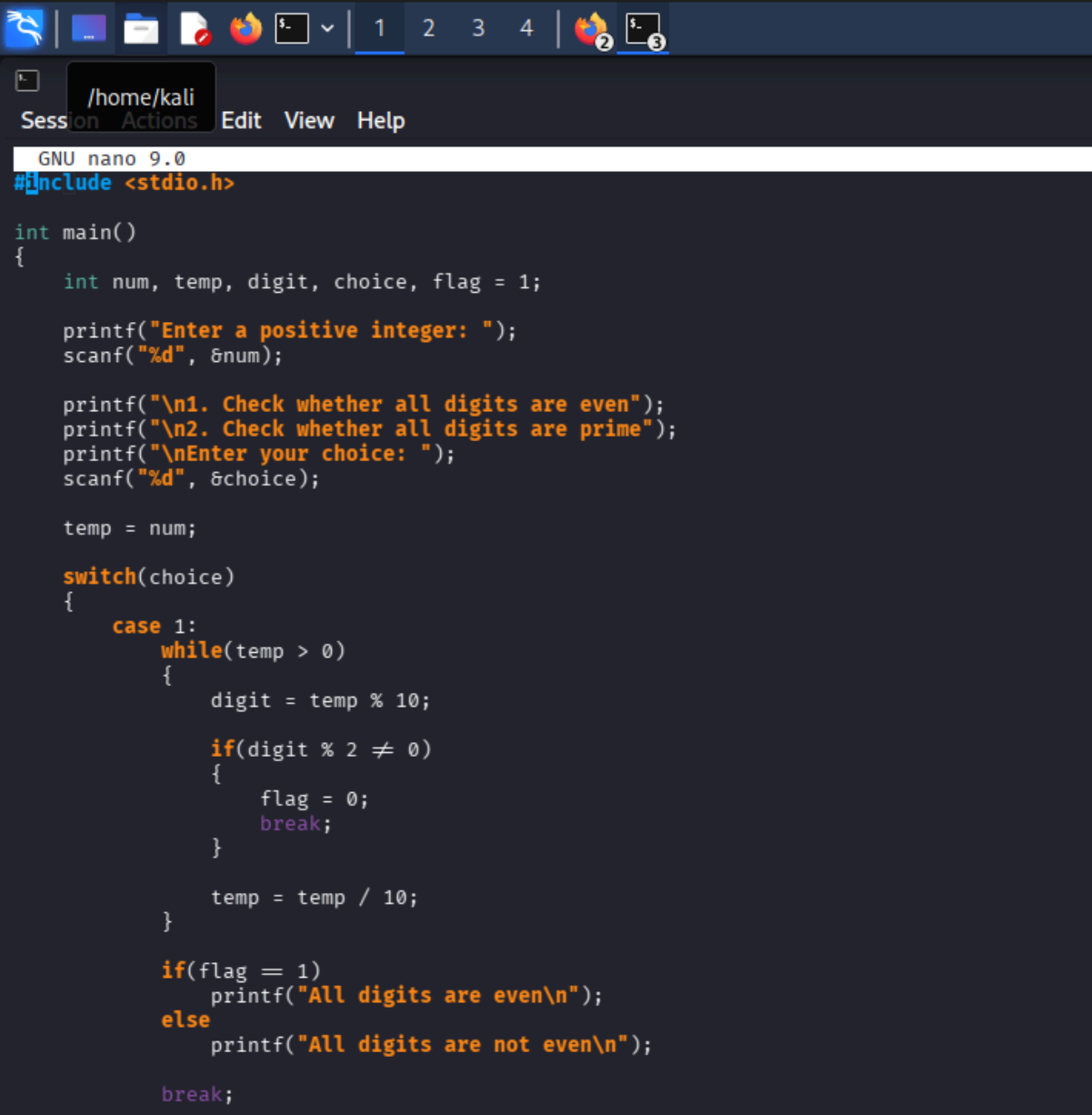
The prime digits are only 2, 3, 5, and 7.

1. Set flag = 1.
2. Repeat while temp > 0:
Find the last digit using digit = temp % 10.
Check whether the digit is 2, 3, 5, or 7.
If it is not one of these digits, set flag = 0 and break the loop.
Otherwise, remove the last digit using temp = temp / 10.
3. If flag == 1, display "All digits are prime".
4. Otherwise, display "All digits are not prime".

2. Test Case Table

Test Case	Number	Choice	Expected Output	Actual Output	Result
1	12345	1	All digits are not even	All digits are not even	Pass
2	24680	1	All digits are even	All digits are even	Pass
3	22468	1	All digits are even	All digits are even	Pass
4	2357	2	All digits are prime	All digits are prime	Pass
5	2379	2	All digits are not prime	All digits are not prime	Pass
6	25327	2	All digits are prime	All digits are prime	Pass

3. Program, Compile and Execution:



```

/home/kali
Session Actions Edit View Help
GNU nano 9.0
#include <stdio.h>

int main()
{
    int num, temp, digit, choice, flag = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    printf("\n1. Check whether all digits are even");
    printf("\n2. Check whether all digits are prime");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    temp = num;

    switch(choice)
    {
        case 1:
            while(temp > 0)
            {
                digit = temp % 10;

                if(digit % 2 != 0)
                {
                    flag = 0;
                    break;
                }

                temp = temp / 10;
            }

            if(flag == 1)
                printf("All digits are even\n");
            else
                printf("All digits are not even\n");

            break;
    }
}

```

```
    case 2:
        while(temp > 0)
        {
            digit = temp % 10;

            if(digit != 2 && digit != 3 &&
               digit != 5 && digit != 7)
            {
                flag = 0;
                break;
            }

            temp = temp / 10;
        }

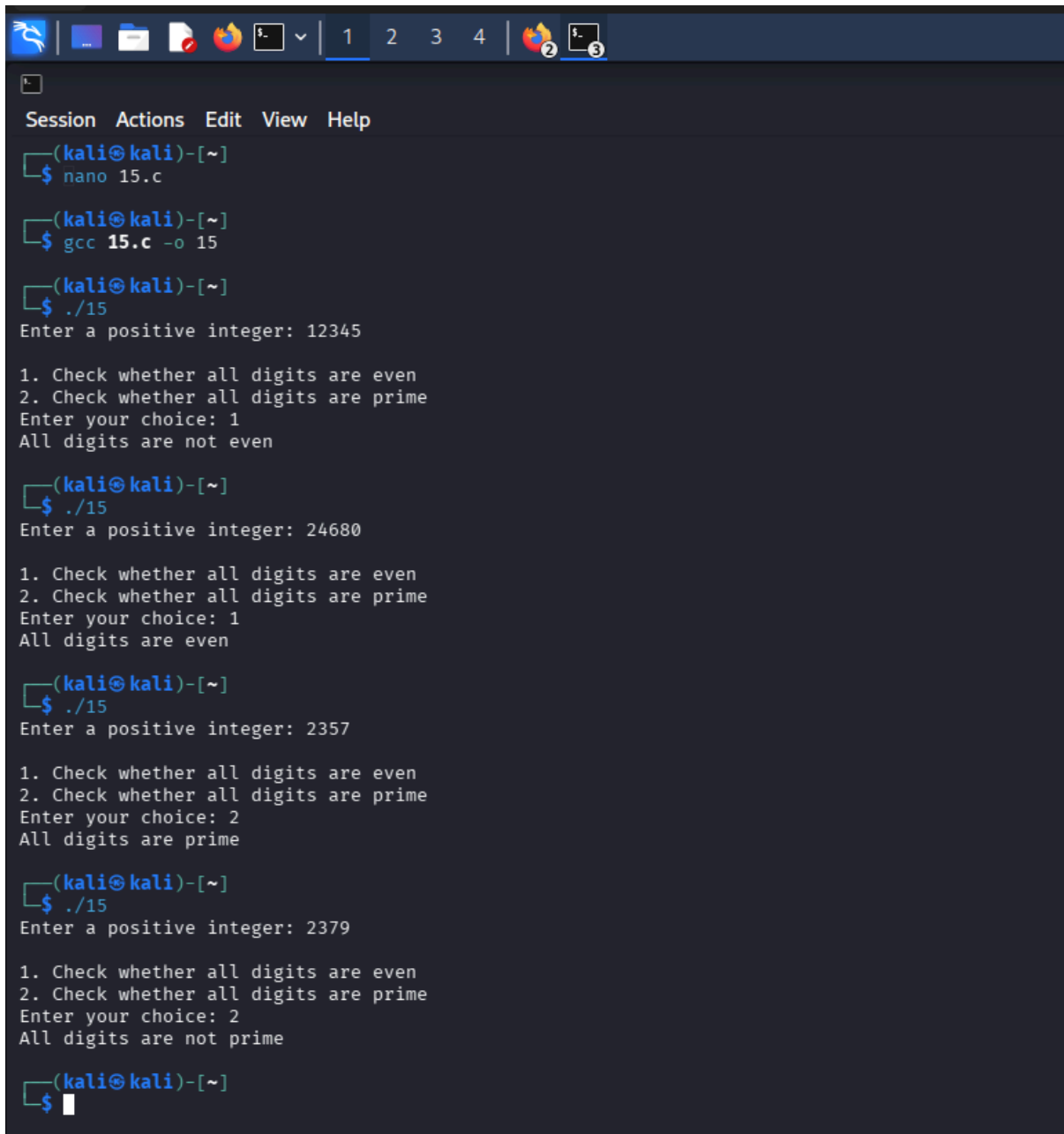
        if(flag == 1)
            printf("All digits are prime\n");
        else
            printf("All digits are not prime\n");

        break;

    default:
        printf("Invalid choice\n");
}

return 0;
```

```
}
```



```
(kali㉿kali)-[~]
$ nano 15.c

(kali㉿kali)-[~]
$ gcc 15.c -o 15

(kali㉿kali)-[~]
$ ./15
Enter a positive integer: 12345

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are not even

(kali㉿kali)-[~]
$ ./15
Enter a positive integer: 24680

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even

(kali㉿kali)-[~]
$ ./15
Enter a positive integer: 2357

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime

(kali㉿kali)-[~]
$ ./15
Enter a positive integer: 2379

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are not prime

(kali㉿kali)-[~]
$
```

4. Conclusion:

The program was successfully written, compiled, executed, and tested. It correctly uses a switch-case menu, loops through the digits of the entered number, and produces the expected results for both operations.

NAME: TAANISH AGARWAL

BATCH: B-15

SAP ID: 590034662